LONGITUDINAL ELECTROMAGNETIC STRESS IN CURRENT-CARRYING CONDUCTORS: AN HQIV/O-MAXWELL EXPLANATION OF AXIAL WIRE FORCES

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Abstract. **Background.** A recent Cambridge Open Engage working paper reports direct measurements of an axial expansion force in metallic wires carrying direct current and argues that the residual force is not explained by thermal expansion, magnetic pinch, or electromigration alone. The result is not yet peer reviewed, but it reopens a long-standing question: whether a current-carrying conductor can support a longitudinal electromagnetic stress parallel to the current.
Purpose. This note gives an HQIV/O-Maxwell account of such a force without discarding the Lorentz force law for individual charges. The key distinction is between the transverse force on a free carrier and the longitudinal stress stored in a constrained, resistive, many-body conductor. In HQIV, the axial term is the spatial component of the same inhomogeneous O-Maxwell source balance already used for currents and phase-gradient corrections.
Result. For a straight conductor with axial coordinate s, cross-sectional area A, current I, conductivity σ, mobile charge density nq, and HQIV scalar φ, the predicted longitudinal electric/stress channel is

$$E_\parallel^{\mathrm{eff}} = \frac{I}{\sigma A} + E_*\,\frac{\alpha}{4\pi}\,\Lambda_s\,\partial_s\varphi, \qquad f_\parallel = nq\,E_\parallel^{\mathrm{eff}},$$

where $\alpha = 3/5$, Λ_s is the algebra-first O-Maxwell coupling log for the axial component, and E_* converts the dimensionless O-Maxwell cell readout to SI. Equivalently, the measurable axial excess over calibrated Ohmic/thermal response is

$$\Delta f_\parallel = nq\,E_*\,\frac{3}{20\pi}\,\Lambda_s\,\partial_s\varphi.$$

The model predicts reversal with current direction only through the current-induced odd part of $\partial_s\varphi$, scaling with the resistive phase-gradient profile rather than with the magnetic pinch pressure $B^2/(2\mu_0)$. This gives a falsifiable separation from heating, pinch, and electromigration.

1. Scope and Claim Discipline

The experimental prompt for this note is the Cambridge Open Engage working paper "Direct Measurement of Longitudinal Electromagnetic (EM) Forces in Wires" [?]. That posting reports axial wire forces under DC current and states that the measurement protocol separates the force from thermal expansion, magnetic pinch, and electromigration. Because the source is a working paper rather than a peer-reviewed article, the present note treats the result as a target phenomenon to be explained and tested, not as an established fact.

The accepted textbook statement is narrower than it is often presented. For a point charge with velocity $\mathbf{v}$,

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v}\times\mathbf{B}),$$

so the magnetic part is perpendicular to $\mathbf{v}$. This does not imply that a resistive many-body conductor cannot carry a longitudinal electromagnetic stress. A conductor is not a free particle: it has lattice constraints, boundary conditions, charge relaxation, contact potentials, Joule heating, and internal momentum exchange between carriers, phonons, surfaces, and fields. The question is therefore not whether $q\mathbf{v}\times\mathbf{B}$ has an axial component for one carrier. It does not. The question is

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whether the field-current-lattice system has an axial stress component after the carrier momentum budget is coarse-grained.

HQIV answers yes, conditionally: a longitudinal component appears when the current drives a nonzero axial phase/lapse gradient in the O-Maxwell source balance. The term vanishes in the flat, constant-φ, uniform-readout limit, so the model preserves the ordinary Maxwell/Lorentz limit.

2. Accepted Mechanisms and the Residual

Let the wire axis be s. In the conventional budget, the main axial readouts are:

$$E_{\mathrm{Ohm}} = \frac{J}{\sigma} = \frac{I}{\sigma A}, \qquad P_{\mathrm{Joule}} = J E_{\mathrm{Ohm}} = \frac{J^2}{\sigma},$$

plus thermal expansion through the temperature field $T(s, r, t)$. The magnetic self-field of a long round wire is azimuthal,

$$B_\theta(r) = \frac{\mu_0 I r}{2\pi R^2} \quad (r < R),$$

and produces a radial pinch stress, not a leading axial stress in an ideal straight isotropic cylinder. Electromigration produces axial momentum transfer, but it is tied to carrier drift, defects, and material transport; it should show strong dependence on microstructure, temperature, and long-time mass displacement.

The reported residual is interesting only if the following calibrated model cannot account for it:

$$F_{\mathrm{meas}}(I,t) = F_{\mathrm{thermal}}(I^2,t) + F_{\mathrm{pinch}}(I^2,t) + F_{\mathrm{electromigration}}(I,t;\mathrm{material}) + F_{\mathrm{residual}}(I,t).$$

The HQIV claim concerns F_{residual}. It does not deny the first three terms.

3. O-Maxwell Source Balance

The relevant formal layer in the Lean corpus is the O-Maxwell action and continuum closure:

$$\mathrm{EL}_{a\nu} = \sum_\mu F_{a\mu\nu} - 4\pi J_{a\nu} - \mathbf{1}_{a=0}\, \alpha \log(\varphi+1)\, \partial_\nu \varphi,$$

with the continuum version replacing $\partial_\nu\varphi$ by chart gradient components or by a metric-raised gradient. In the algebra-first Maxwell layer, the same correction is written with the component coupling $\log \Lambda_\nu$:

$$\mathcal{M}_{a\nu} = -4\pi J_{a\nu} - \alpha\, \Lambda_\nu\, \partial_\nu \varphi,$$

up to the common divergence side of the equation.

For the electromagnetic channel $a = 0$ and axial component $\nu = s$, stationarity gives

$$\sum_\mu F_{0\mu s} = 4\pi J_{0s} + \alpha\, \Lambda_s\, \partial_s \varphi.$$

Thus the source that a coarse-grained conductor reads along its axis is

$$J_s^{\mathrm{eff}} = J_s + \frac{\alpha}{4\pi} \Lambda_s\, \partial_s \varphi.$$

This is the minimal mathematical location of the longitudinal term. It is not an extra force law pasted onto Lorentz theory; it is the longitudinal component of the same inhomogeneous source equation once the HQIV phase-gradient slot is nonzero.

4. Force Density

The carrier/lattice force density associated with an effective axial field is

$$f_\parallel = nq\, E_\parallel^{\mathrm{eff}},$$

where n is the mobile carrier number density and q is the carrier charge magnitude. In SI readout,

$$E_\parallel^{\mathrm{eff}} = E_{\mathrm{Ohm}} + E_{\mathrm{HQIV}}, \qquad E_{\mathrm{Ohm}} = \frac{I}{\sigma A},$$

and

$$E_{\mathrm{HQIV}} = E_* \frac{\alpha}{4\pi} \Lambda_s\, \partial_s \varphi = E_* \frac{3}{20\pi} \Lambda_s\, \partial_s \varphi.$$

Here E_* is not a fitted material constant. It is the unit/readout conversion between the dimensionless O-Maxwell cell residual and SI electric field units. In a purely natural-unit HQIV computation, $E_* = 1$.

For a wire segment of length L and cross-section A, the excess axial force is

$$\Delta F_\parallel = A \int_0^L nq\, E_{\mathrm{HQIV}}(s)\, ds = A\, nq\, E_* \frac{3}{20\pi} \int_0^L \Lambda_s(s)\, \partial_s \varphi(s)\, ds.$$

If Λ_s is approximately constant over the segment,

$$\Delta F_\parallel \approx A\, nq\, E_* \frac{3}{20\pi} \Lambda_s \big[\varphi(L) - \varphi(0)\big].$$

This boundary form is experimentally important: a uniform interior with symmetric contacts can suppress the integrated signal even if local stresses exist; asymmetric contacts, gradients, or clamps can reveal it.

5. Connection to Phase-Horizon Tipping

The algebra-first O-Maxwell seed defines the phase-horizon tipping angle

$$\delta\theta'(E') = \arctan(E')\, \frac{\pi}{2},$$

and the effective fluid closure uses the vacuum momentum source

$$\mathbf{g}_{\mathrm{vac}} = -\frac{\gamma}{6} \nabla(\varphi\, \dot{\delta\theta'}).$$

For a conductor in steady current, the axial component is

$$g_{\mathrm{vac},s} = -\frac{\gamma}{6} \left(\varphi\, \partial_s \dot{\delta\theta'} + \dot{\delta\theta'}\, \partial_s \varphi \right), \qquad \gamma = \frac{2}{5}.$$

This gives the mechanical stress channel complementary to the O-Maxwell source equation. The source-balance expression identifies the longitudinal electromagnetic residual; the fluid expression says how that residual appears as momentum transfer into the lattice/continuum medium.

At small electric proxy E',

$$\delta\theta'(E') = \frac{\pi}{2} E' + O((E')^3).$$

Since E' is driven by local electric energy/readout, the leading current dependence of the HQIV axial term is determined by which part of $\partial_s \varphi$ and $\partial_s \dot{\delta\theta'}$ is odd or even under $I \mapsto -I$. This is a discriminator:

$$F_{\mathrm{thermal}}, F_{\mathrm{pinch}} \sim I^2, \qquad F_{\mathrm{HQIV}} \sim c_1 I + c_2 I^2 + c_3 I^3 + \cdots,$$

with the odd part controlled by contact/asymmetry-induced phase gradients.

6. Why This Does Not Contradict the Lorentz Force

The standard objection is:

$$\mathbf{v} \cdot (\mathbf{v} \times \mathbf{B}) = 0,$$

so magnetic force cannot do work along the carrier motion. HQIV agrees with that statement. The proposed longitudinal term is not the axial projection of $\mathbf{v} \times \mathbf{B}$. It is a constrained-medium source term:

$$\partial_\mu F^{\mu s} = 4\pi J^s + \alpha \Lambda_s \partial^s \varphi.$$

The additional axial stress is carried by the phase/lapse gradient and then exchanged with the lattice through the same current source budget. In the constant-φ limit,

$$\partial_s \varphi = 0, \qquad \partial_s \delta\dot{\theta}' = 0,$$

and the longitudinal residual vanishes. The accepted meta is therefore recovered as a limiting case, not rejected.

7. Order-of-Magnitude Readout

For copper-like conductivity $\sigma \approx 5.8 \times 10^7\,\mathrm{S/m}$, current $I = 10\,\mathrm{A}$, and radius $R = 1\,\mathrm{mm}$,

$$A = \pi R^2 \approx 3.14 \times 10^{-6}\,\mathrm{m}^2, \qquad J \approx 3.18 \times 10^6\,\mathrm{A/m}^2,$$

so

$$E_{\mathrm{Ohm}} \approx 5.5 \times 10^{-2}\,\mathrm{V/m}.$$

The dimensionless fractional residual is

$$\epsilon_{\mathrm{HQIV}} := \frac{E_{\mathrm{HQIV}}}{E_{\mathrm{Ohm}}} = \frac{E_*\,(3/20\pi)\Lambda_s \partial_s \varphi}{I/(\sigma A)}.$$

This form is useful because it does not require guessing the absolute HQIV readout scale before doing comparisons. Experiments can first fit or bound ϵ_{HQIV} after subtracting the calibrated thermal/pinch/electromigration terms, then test whether the inferred value scales with $\partial_s\varphi$-like controls: contact asymmetry, material coherence, current density profile, and shell/temperature gradients.

8. Experimental Discriminants

The HQIV residual differs from standard backgrounds in the following ways.

(1) **Current reversal.** Thermal expansion and magnetic pressure are even in I. A phase-gradient residual can contain an odd component if the conductor/contact geometry creates an oriented $\partial_s \varphi$ or $\partial_s \delta\dot{\theta}'$. Measure

$$F_{\mathrm{odd}}(I) = \frac{F(I) - F(-I)}{2}, \qquad F_{\mathrm{even}}(I) = \frac{F(I) + F(-I)}{2}.$$

(2) **Contact dependence.** If the integrated force is approximately a boundary term,

$$\Delta F_{\parallel} \propto \varphi(L) - \varphi(0),$$

then changing contacts, clamps, surface preparation, or lead geometry should alter the residual more strongly than it alters bulk Joule heating.

(3) **Length dependence.** A uniform bulk thermal expansion force scales differently from a boundary-dominated phase-gradient force. Under fixed I and radius, compare multiple L while independently measuring temperature fields.

(4) **Pulse timing.** Thermal expansion follows heat diffusion times. The O-Maxwell source response should track electrical/phase equilibration first, with thermal drift later. Fast pulses can separate these channels.

(5) **Material coherence.** The fluid closure predicts sensitivity to coherence and microstructure through the effective medium terms. Annealed, cold-worked, polycrystalline, and single-crystal samples should not collapse to one universal thermal curve if the residual is real.

9. Minimal Test Equation

For data analysis, use

$$F_{\text{meas}}(I,t) = a_2(t)I^2 + a_4(t)I^4 + b_1(t)I + b_3(t)I^3 + F_0(t),$$

where a_2, a_4 absorb thermal and magnetic even-in-current channels, while b_1, b_3 capture oriented longitudinal residuals. HQIV predicts that the odd coefficients should be correlated with axial phase-gradient controls:

$$b_1, b_3 \longleftrightarrow \frac{3}{20\pi} A\, nq\, E_* \int \Lambda_s\, \partial_s\varphi\, ds \quad \text{and} \quad -\frac{\gamma}{6}\int \partial_s(\varphi\delta\dot{\theta}')\, ds.$$

If all odd components vanish under carefully reversed contacts and the remaining even residual scales exactly with measured temperature and pinch pressure, the HQIV explanation is not needed. If an oriented component persists after those controls, the O-Maxwell phase-gradient channel is the natural place in the present framework to put it.

10. Astrophysical extension: coronal magnetic flux tubes

The three preconditions for the longitudinal residual to be non-negligible (axial current channel, non-zero $\partial_s\varphi$, asymmetric boundary geometry) are jointly extremized in solar coronal magnetic loops: photospheric footpoints provide opposite-polarity asymmetric “contacts,” the chromosphere–transition region–corona stratification realises the steepest φ gradient in the solar atmosphere via the HQIV shell ladder $\varphi(m) = 2(m+1)$, and the loop's force-free flux tube is the natural current-carrying conductor. The boundary form

$$\frac{Q_\parallel}{A} = nq\, v_\parallel\, E_*\, \frac{3}{20\pi}\, \Lambda_s \left[\varphi(\text{corona}) - \varphi(\text{photosphere})\right]$$

gives a candidate *third* steady-state coronal-heating channel beside Alfvén-wave dissipation and Parker-type nanoflare reconnection. The companion note [?] works out the discriminants (footpoint-polarity asymmetry, length-independence, sub-Alfvénic timing, network correlation) and the Lean module `Hqiv/Physics/CoronalLongitudinalStress.lean` [?] registers each equation as a `Prop` theorem with zero `sorry`.

11. Conclusion

The accepted meta says that the magnetic part of the Lorentz force is transverse to carrier velocity. That statement is correct but incomplete for a constrained resistive conductor. HQIV/O-Maxwell predicts a longitudinal conductor stress when the current establishes an axial phase/lapse gradient. The quantified residual is

$$\Delta F_\parallel = A\, nq\, E_*\, \frac{3}{20\pi} \int_0^L \Lambda_s\, \partial_s\varphi\, ds,$$

with a companion momentum-source expression

$$g_{\text{vac},s} = -\frac{1}{15}\left(\varphi\, \partial_s\delta\dot{\theta}' + \delta\dot{\theta}'\, \partial_s\varphi\right),$$

using $\gamma = 2/5$. Both vanish in the constant-φ, symmetric, flat Maxwell limit. The theory therefore explains a possible longitudinal wire force as a residual phase-gradient stress, not as a violation of the point-particle Lorentz force law.

References

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